COVID Mortality Disruption Lab

Did COVID-19 disrupt the trend for six other leading causes of death, has it recovered, and where did it hit hardest?

Aidan Chi · October 2026

Abstract

Background. Deaths attributed to COVID-19 understate the pandemic's effect on mortality, because disrupted care and other indirect effects also change death rates from other causes.

Methods. Using CDC WONDER age-adjusted death rates for 1999–2024, I projected each of six leading causes' pre-pandemic linear trend with a 95% prediction interval and tested whether 2020–21 rates departed from it. Predictions were written down before any 2020–2024 data were analyzed. A negative control, four sensitivity analyses, false discovery rate (FDR) correction and autocorrelation-robust standard errors test the results. County changes from 2015–19 to 2020–24 were regressed on five County Health Rankings measures.

Results. Five of six causes rose significantly above trend in 2020–21, led by drug overdose (+40.9%) and diabetes (+26.9%); all survive FDR correction. By 2024, diabetes (+15.0%) and cancer (+4.6%) remained above trend and Alzheimer's disease (−19.1%) had fallen below it; heart disease, stroke, and drug overdose were back within their normal ranges. The 2024 status of four causes depends on the baseline window, and the 2020–21 results for stroke and cancer depend on the trend's shape. Across counties, higher smoking and uninsured rates were associated with larger increases for all five disrupted causes.

Conclusions. The pandemic's indirect effect on these causes was concentrated in drug overdose and diabetes and had mostly faded nationally by 2024. Where gaps remain, their size depends heavily on how the no-pandemic trend is modeled, so they are best read as departures from a projected trend rather than as changes from 2019 levels.

1. Introduction

Counts of deaths attributed to COVID-19 capture only part of the pandemic's effect on mortality. Delayed or interrupted medical care, strained hospitals, isolation and economic stress can also change death rates from other causes, and those deaths are recorded under those causes, not under COVID-19. Excess-mortality analysis, which compares observed deaths with what earlier trends predicted, is the standard way to measure that wider effect.

This study applies that approach to six leading causes of death in the United States: heart disease, diabetes, Alzheimer's disease, stroke, drug overdose and cancer. It asks whether each cause departed from its pre-pandemic trend in 2020–21, whether it had returned to trend by 2024, and where across U.S. counties the changes were largest. The expected direction and confidence for each cause were recorded before any 2020–2024 data were analyzed, so the results can be judged against stated predictions rather than explained after the fact. COVID-19 itself is treated as the shock, not as an outcome. Mortality data can show what changed, by how much and where, but not why: direct infection, delayed care and economic stress leave the same mark, so every result is an association.

2. Data and methods

2.1 Data

National age-adjusted death rates (2000 U.S. standard population) come from CDC WONDER's Underlying Cause of Death files: the 1999–2020 database for 1999–2019 and the 2018–2024 single-race database for 2020–2024. In 2018–19, the years both databases cover, every cause matched exactly. Table 1 lists the causes, their ICD-10 codes and the predictions made beforehand. County data are pooled five-year totals for 2015–19 and 2020–24, which CDC reports only as crude rates, matched to five measures from the 2024 County Health Rankings & Roadmaps release.

Table 1. Causes analyzed.

Cause	ICD-10	Baseline	Prediction made beforehand
Drug overdose	X40-X44, X60-X64, X85, Y10-Y14	1999–2019	Largest disruption; may reverse by 2023–24
Diabetes	E10-E14	1999–2019	Real disruption; persistence uncertain
Heart disease	I00-I09, I11, I13, I20-I51	2010–2019	Real disruption; persistence uncertain
Alzheimer's disease	G30	1999–2019	Large, isolation-driven
Stroke	I60-I69	2010–2019	Real but harder to detect
Cancer	C00-C97	1999–2019	No disruption expected yet
Congenital malformations	Q00-Q99	1999–2019	Negative control: no disruption

2.2 Expected trend and disruption test

For each cause, an ordinary least squares line was fit to its pre-pandemic rates and projected through 2024 with a 95% prediction interval, which reflects both year-to-year variation and uncertainty in the fitted line. A year counts as disrupted when its observed rate falls outside that interval. The primary test compares the average 2020–21 gap with its exact standard error, which counts the line's uncertainty once and averages only the year-to-year noise. The breakpoint is fixed at 2020 rather than searched for. Heart disease and stroke use a 2010–2019 baseline: using only pre-2020 data, a straight line over 1999–2019 already sat well below their flattened 2019 rates (Section 3.4).

Causes with a significant 2020–21 disruption are classed by what happened in 2022–24: *Persisted* if any later year stayed significantly off trend in the same direction, *Reversed* if in the opposite direction, and *Resolved* if none did.

2.3 Checks on the method

- **Negative control.** The same pipeline was run on congenital malformations, which have no direct link to COVID-19, using death counts because the rate is reported to one decimal at about 3 per 100,000.
- **Multiple testing.** Benjamini-Hochberg FDR correction across the six causes, and separately across the five county measures within each cause [7].
- **Autocorrelation.** The primary test assumes independent baseline years; a Newey-West version allows for correlation between neighboring years [8, 9].
- **Sensitivity analyses.** The 2020–21 test was rerun with the other baseline window, a stricter significance level ($\alpha = 0.01$), a curved (quadratic) trend, and a trend that changes by a steady percentage each year (linear in log rate).
- **Cross-check and full-period test.** Three change-point methods search each series for a break without being told about 2020, and a secondary test pools all five post-2020 years to catch delayed effects.

2.4 County analysis

For each of the five causes with a 2020–21 disruption, each county's change is its 2020–24 crude rate minus its 2015–19 rate. A county is included when CDC reports a reliable rate in both periods (about 20 deaths over five years). The change was regressed separately on the uninsured rate, adult smoking, adult obesity, median household income and the share of residents living in rural areas. Alzheimer's disease, which had no 2020–21 disruption, was run the same way as an exploratory comparison.

3. Results

3.1 Disruption in 2020–21

Five of the six causes rose significantly above their pre-pandemic trend in 2020–21, and all survive FDR correction (Table 2). The largest increases were for drug overdose and diabetes. Figure 1 shows the method on drug overdose: deaths broke above the prediction interval in 2020 and stayed there through 2023 before falling back. Significance and size are different things. Cancer's 2020–21 gap is only 1.7% but significant because its year-to-year variation before 2020 was so small, while overdose needs a much larger gap to stand out (Figure 2).

Table 2. Departure from the pre-pandemic trend, by cause.

Cause	Classification	2020–21 gap	p-value	FDR	2024 gap	Trend shape
Drug overdose	Persisted	+40.9%	2.1×10^{-7}	Yes	–4.3%	Robust
Diabetes	Persisted	+26.9%	2.5×10^{-6}	Yes	+15.0%	Robust
Heart disease	Persisted	+7.8%	1.1×10^{-4}	Yes	+3.0%	Robust
Cancer	Persisted	+1.7%	4.5×10^{-4}	Yes	+4.6%	Sensitive
Stroke	Persisted	+8.8%	0.004	Yes	+6.1%	Sensitive
Alzheimer's disease	No significant disruption	+1.0%	0.82	No	–19.1%	Robust

Gaps are observed minus expected, as a percent of expected. p-values are for the 2020–21 average. Trend shape: whether the result holds with a curved and with a steady-percentage trend (Section 3.4).

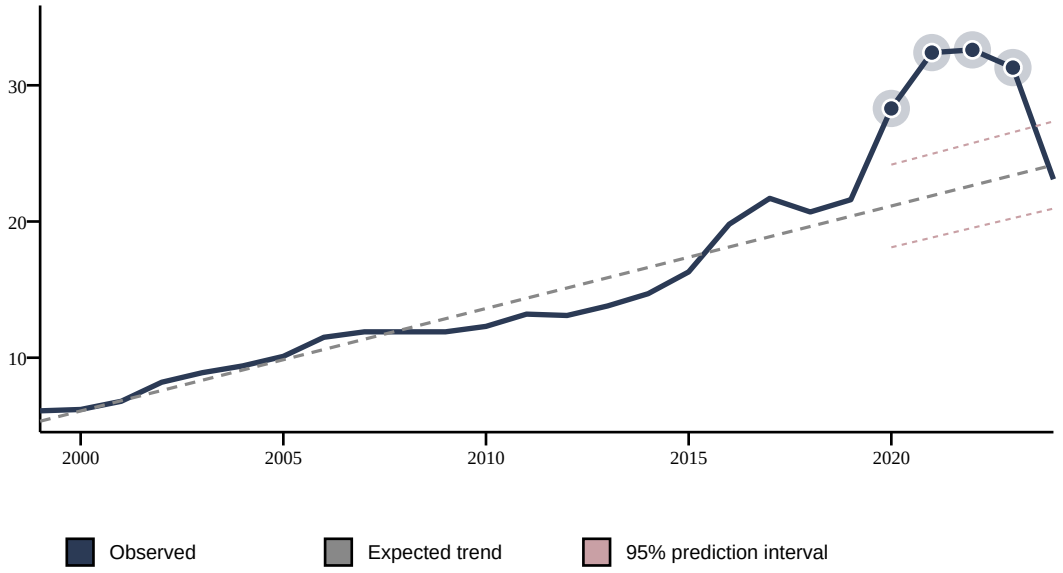

Figure 1. Drug overdose death rate against its 1999–2019 trend. The dashed line is the trend; the pink lines mark its 95% prediction interval; circled points fall outside it.

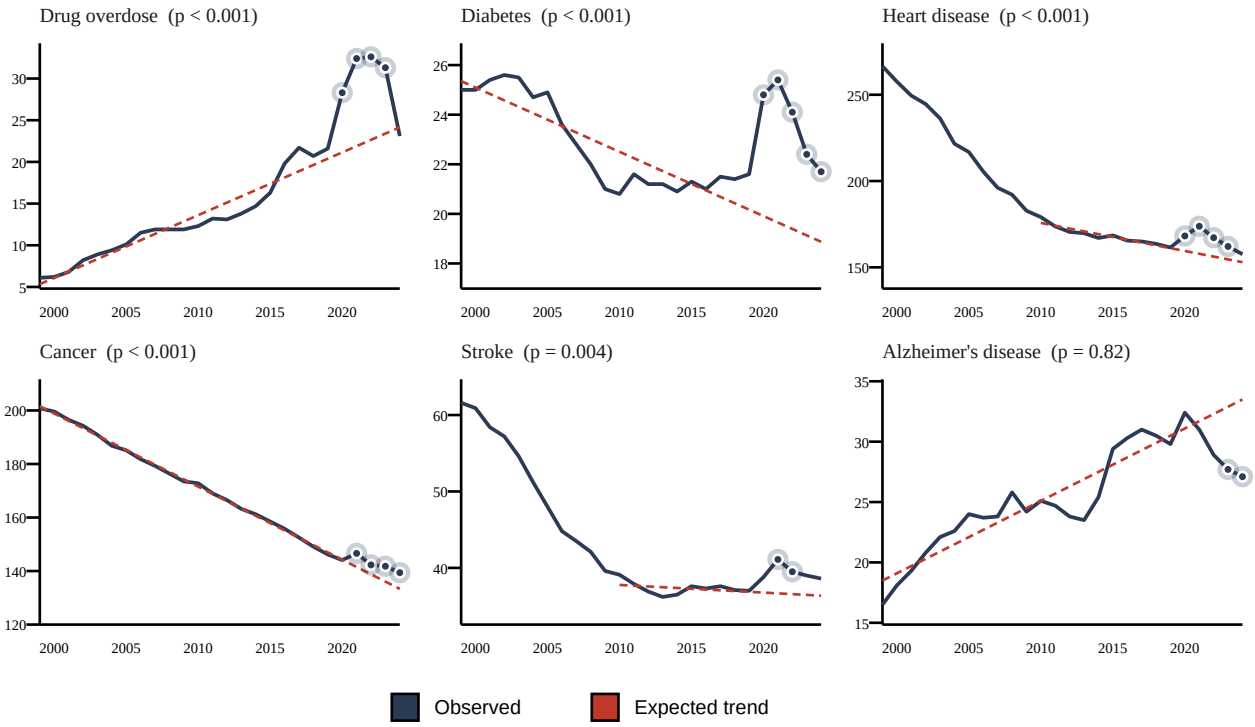

Figure 2. All six causes, observed rate against the pre-pandemic trend, 1999–2024. Circled points fall outside the 95% prediction interval. Panel titles give the 2020–21 p-value.

3.2 Where each cause stood in 2024

Most of the disruption had faded by 2024 (Figure 3). Diabetes (+15.0%) and cancer (+4.6%) were still significantly above trend, and Alzheimer's disease (–19.1%) had fallen significantly below it. Heart disease, stroke, and drug overdose were back within their normal ranges. Drug overdose fell from its peak to –4.3%, slightly below trend but not significantly.

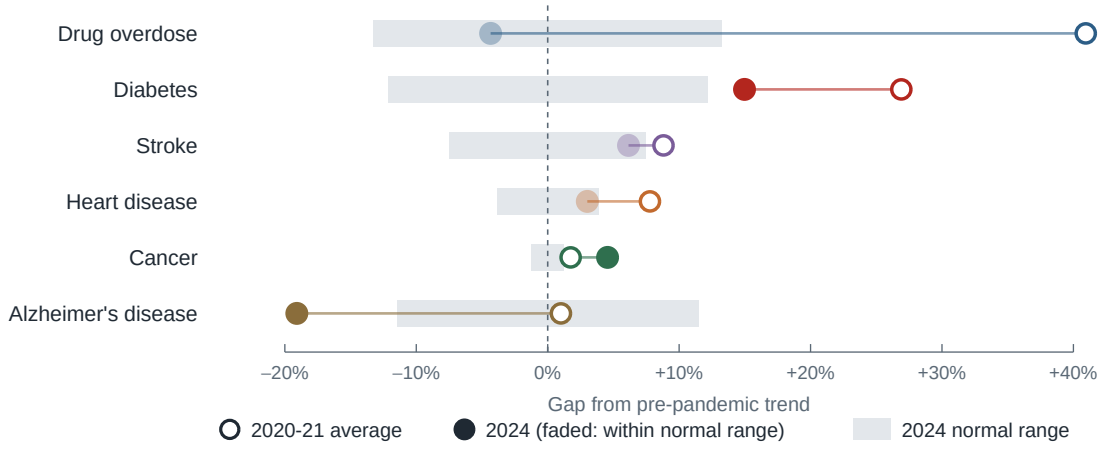

Figure 3. Gap from trend in 2020–21 and 2024. Gray band: 2024 95% prediction interval; a gap outside it is significant.

"Above trend" is not the same as "above 2019." Figure 4 compares each cause's actual change since 2019 with the change its pre-2020 trend projected. Diabetes's 2024 rate was back at its 2019 level (21.7 vs. 21.6); it reads as above trend because its trend line was still falling. Stroke is back within its normal range but still 4% above its 2019 rate. Drug overdose is back within its normal range but still 7% above its 2019 rate. Cancer kept falling (146.2 in 2019, 139.4 in 2024), just more slowly than its trend.

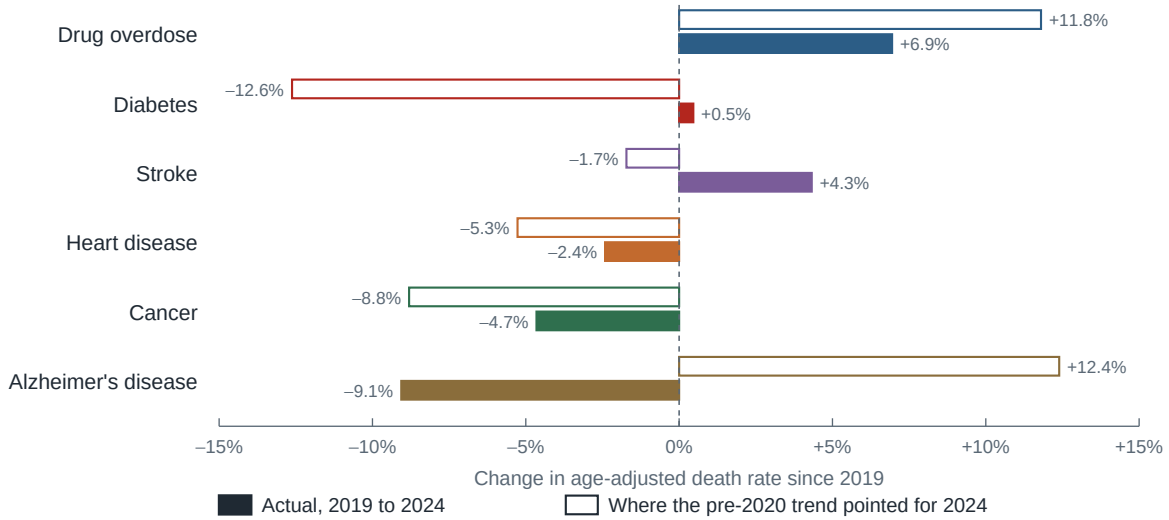

Figure 4. Change in each cause's age-adjusted death rate from 2019 to 2024: actual (filled) and as projected by the pre-2020 trend (outlined). The further apart the two bars, the larger the 2024 gap from trend.

3.3 Results that went against the predictions

Cancer. No disruption was expected this soon, since missed screenings were thought to take longer to show up as deaths. Instead cancer rose 1.7% above trend in 2020–21 ($p < 0.001$) and was 4.6% above it in 2024. The result is fragile: with a steady-percentage trend, the 2020–21 gap is no longer significant.

Alzheimer's disease. A large increase was expected from isolation and disrupted care. The 2020–21 test found none ($p = 0.82$): 2020 ran 4.2% above trend, but 2021 fell below it. Pooling all five post-2020 years does find a significant later decline ($p = 0.016$), reaching –19.1% by 2024.

3.4 Robustness

Baseline correction. Heart disease and stroke originally used the 1999–2019 baseline. Using only pre-2020 data, an F-test showed strong curvature for these two causes ($F = 222.7$ and 162.6, both $p < 0.00001$) and not for the others ($F < 9.5$): both fell steeply through the 2000s and then leveled off, so the full-period line sat 17.7 and 5.8 points below their actual 2019 rates. A curved fit would project rising rates after 2019, so a straight line over 2010–2019 is used instead. Both stay significant, with 2020–21 gaps of 8–9% rather than 26–37%.

Sensitivity analyses. Table 3 shows each check. Drug overdose, diabetes, and heart disease hold under all of them. Stroke loses significance with a curved trend ($p = 0.18$); cancer loses significance with a steady-percentage trend ($p = 0.43$). Stroke also changes classification at the stricter significance level, making stroke and cancer the two least certain results. Every significant cause stays significant with autocorrelation-robust standard errors. The negative control shows no disruption ($p = 0.71$ on death counts).

Table 3. Does the 2020–21 result hold under each alternative?

Cause	2020–21 result	Other window	$\alpha = 0.01$	Curved trend	Steady-% trend	Robust p-value
Drug overdose	Significant	Holds	Holds	Holds	Holds	2.5×10^{-5}
Diabetes	Significant	Holds	Holds	Holds	Holds	6.7×10^{-5}
Heart disease	Significant	Holds	Holds	Holds	Holds	2.7×10^{-5}
Cancer	Significant	Holds	Holds	Holds	Changes	6.6×10^{-4}
Stroke	Significant	Holds	Changes	Changes	Holds	0.0021
Alzheimer's disease	Not significant	Holds	Holds	Holds	Holds	0.86

Holds: same classification as the primary analysis. Robust p-value: Newey-West (autocorrelation-robust) version of the 2020–21 test.

2024 status and the baseline window. The 2020–21 result holds under either baseline window for every cause, but the 2024 status is more sensitive (Table 4). Against a 2010–2019 trend, diabetes would be back within its normal range and drug overdose significantly below trend. For heart disease and stroke, the other window is the 1999–2019 line already shown to misfit them.

Table 4. 2024 status under each baseline window.

Cause	Primary baseline	2024 gap	Other baseline	2024 gap
Heart disease	2010–2019	+3.0% (within)	1999–2019	+34.9% (above)
Diabetes	1999–2019	+15.0% (above)	2010–2019	+0.2% (within)
Alzheimer's disease	1999–2019	–19.1% (below)	2010–2019	–23.9% (below)
Stroke	2010–2019	+6.1% (within)	1999–2019	+57.3% (above)
Drug overdose	1999–2019	–4.3% (within)	2010–2019	–18.0% (below)
Cancer	1999–2019	+4.6% (above)	2010–2019	+5.4% (above)

3.5 Variation across counties

The county analysis covers 1,564 to 3,066 counties per cause (3,066 for heart disease, 1,564 for drug overdose). Higher smoking and uninsured rates went with larger increases for all five causes; higher household income went with smaller increases for four of the five causes; higher obesity rates went with larger increases for three of the five causes. The link with rurality was mixed: larger for three causes, smaller for one (Table 5).

Table 5. Association between county measures and the change in death rate, 2015–19 to 2020–24.

County measure	Heart	Diabetes	Stroke	Overdose	Cancer
% uninsured	+77.9	+15.2	+15.3	+16.8	+35.5
% adult smokers	+199.5	+49.7	+26.9	+84.6	+40.2
% adult obesity	+130.0	+35.6	+8.9	+40.2	+15.6
Income (per $10k)	–4.1	–1.2	–0.3	–2.1	–0.8
% rural	+13.6	+1.0	–3.2	+2.3	+7.3

Slopes: change in deaths per 100,000 for a full 0–1 change in a share, or per $10,000 of income. Orange: significantly larger increase; blue: significantly smaller; gray: not significant after FDR correction within each cause.

Rurality. An earlier version of the county analysis required a reliable rate in individual years, which kept only 926 counties for diabetes and 641 for drug overdose, mostly larger and less rural ones, and pointed to smaller disruption in rural counties. With five-year totals that pattern disappears for diabetes and reverses for drug overdose. Counties still left out are mostly rural (83–94% rural, against 45–63% for included counties), and county rates are not age-adjusted, so a rural association can partly reflect older populations.

Exploratory: Alzheimer's disease. Run the same way across 2,426 counties, no county measure was significantly associated with its change.

4. Discussion

The national pattern agrees with published figures. Overdose deaths rose sharply in 2020–21 and fell by about a quarter from 2023 to 2024 [2]; by 2024 the heart disease death rate was below its 2019 level and stroke was within 5% of it [1]; and excess deaths among people with dementia were concentrated in the first pandemic year [4]. The rates used here match CDC's published 2022–2024 figures exactly [1, 3].

Two results need careful reading. The first is what "above trend" means: diabetes in 2024 was back at its 2019 rate and reads as elevated only because its pre-2020 trend was still falling, which is also why its status depends on the baseline window. The second is cancer. Its 2020–21 gap is small and depends on the trend's shape, and outside evidence is mixed: modeling of England's actual 2020–22 deaths found breast and lung cancer deaths close to their pre-pandemic trends [6], while a review of pre-pandemic studies links each four-week delay in starting treatment with higher mortality [5]. A small effect is plausible but not established.

The Alzheimer's result reflects the outcome definition as much as the pandemic. This study counts Alzheimer's only as the underlying cause of death, and when a person with dementia died of COVID-19, COVID-19 was usually recorded as the underlying cause. Studies that also count dementia as a contributing cause found about 94,700 excess deaths in the first pandemic year [4]. The later decline found here is consistent with mortality displacement, but these data cannot confirm it.

Across counties, the strongest and most consistent associations were with smoking and the uninsured rate, which went with larger increases for every disrupted cause. These are ecological associations between county averages; they describe where increases were larger, not why.

5. Limitations

- The design is observational and ecological, so it supports no causal claims, and county patterns may not hold for individuals.
- The projected trend is a model. Results depend on the baseline window and trend shape for four causes' 2024 status and for two causes' 2020–21 result; Section 3.4 reports each.
- Four causes have autocorrelated baseline years, which makes the classical prediction intervals somewhat too narrow; the robust p-values address this for the 2020–21 test only.
- Death-certificate coding may have shifted in 2020–21, when some deaths could be recorded as either COVID-19 or another cause.
- County rates are crude, counties with too few deaths are excluded (mostly rural ones), and the county measures come from 2024, during rather than before the pandemic. Smoking and obesity are modeled estimates, and spatial correlation between neighboring counties is not modeled.
- Two changes were made after results were known: the heart disease and stroke baseline, and the switch to pooled county data. Both are documented, and the earlier versions are reported alongside.

6. Data availability

All results use real CDC WONDER exports: 15 national and 18 county files. The full method, the predictions made in advance, every later change and the export steps are documented in the project repository, which also hosts the interactive app: github.com/aidanc667/covid-mortality-disruption-lab. This project uses public aggregate data for research and education and is not medical advice.

References

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